Global Automated Mesh Application

End-to-End GitOps Infrastructure & Orchestration Pipeline

AWS | Terraform | Ansible | Kubernetes | GitHub Actions

Project Summary

A sophisticated automation project demonstrating the seamless integration of Infrastructure as Code (IaC) and Configuration Management. This pipeline handles the complete lifecycle of a cloud environment, from VPC provisioning to application orchestration within a Kubernetes cluster.

Core Capabilities

- **Automated Infrastructure:** Terraform modules define the AWS network topology, security groups, and compute resources.
- **Dynamic Inventory Management:** A custom GitHub Actions workflow captures ephemeral cloud metadata (IP addresses) and feeds them into Ansible.
- **Kubernetes Orchestration:** Automated deployment of K3s and containerized Nginx services using NodePort mapping.
- **State Consistency:** Ensuring the live environment always matches the declarative code in the repository.

Technical Stack

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- Cloud: Amazon Web Services (AWS)
- IaC: Terraform 1.7.0
- Config: Ansible (Playbooks & Dynamic Variables)
- CI/CD: GitHub Actions
- Container: K3s (Lightweight Kubernetes)
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Workflow Architecture

1. **Code Push:** Triggers GitHub Action.
2. **Terraform:** Validates and applies infrastructure state.
3. **Dynamic Handover:** IP outputs are saved to the runner environment.
4. **Ansible:** Executes system hardening and K3s installation.
5. **Deployment:** Kubernetes Pods and Services are initialized.

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